Assignment - 3

⟹ Problem Statement

Write a Java Program to implement method overloading & method overriding in Java.

1) Write a class Shape. find the area of basic shapes using polymorphism.
2) Write a Java program to create a parent class Hillstations & three subclasses by hill station names

⟹ Objective

1) To study method overloading.
2) To study method overriding in Java.

⟹ Theory

1) Method overloading & its advantages.
- Method overloading occurs when a class has multiple methods with the same name but different parameter lists.
- It is resolved during compiletime.
- Advantages

Increases program readability & allows methods to perform similar tasks with different inputs.

2) Method Overriding.

- It occurs when a subclass provides a specific implementation for a method that is already defined in its superclass.
- Rules:
  The method must have the same name, same return type, & the same parameters.
  It is resolved at runtime.

⟹ Algorithm

1) For Overloading :-

- Create a "Shapes" class.
- Define multiple constructors to handle different shape dimensions.
- Implement methods with the same name but different signatures to display results.

2) For Overriding :-

- Define a base class "Hillstations" with generic methods.
- Create three subclasses that provide unique implementations of those methods.

3) Dynamic Dispatch :-

- In the main method, declare a reference variable of the parent class "Hillstations" & assign it objects of the subclasses to trigger runtime polymorphism.

⇒ Conclusion

Successfully demonstrated the two types of polymorphism in Java. We used Method Overloading to handle shape calculations flexibility & Method Overriding to allow specific hill station behaviors to be involved via a general parent reference at runtime.

⇒ FAQs

1) Can we overload the Java main() method?
→ Yes, you can, but the JVM will only call the standard public static void main(String[] args) as the entry point.

2) Can we declare overloaded methods as final?
→ Yes, Overloaded methods can be final. This simply prevents them from being overridden in subclasses.

3) Can an overloaded method be overridden?
→ Yes, If a subclass inherits overloaded methods, it can choose to override any or all of them.

4) What is method overriding?

→ It is a feature that allows a subclass to provide a specific implementation of a method that is already provided by its parent class.

5) What happens if we change the arguments of an overriding method?

→ It is no longer considered overriding, instead, it becomes Method Overloading!

6) Can we change the return type of an overriding method from Number to Integer?

→ Yes, This is known as Covariant Return Type. Since, Integer is a subclass of Number, Java allows this change during overriding.